

Hardwin Bui

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SKILLS

Languages: C++, C#, Python, JavaScript, TypeScript, HTML, CSS, Blueprints, GLSL

Technologies: React, MongoDB, Express.js, Node.js, OpenGL, Debugger/Profiler

Software: Unity Game Engine, Unreal Game Engine, Linux, Git, GitHub, Visual Studio

Methodologies: Agile, Lean, Refactoring, Design Patterns, Unit Testing, SOLID, DRY, KISS, SCRUM

WORK EXPERIENCE

Google

Jan 2025 - Present

Software Engineer

Mountain View, CA

- Maintained and refactored a core Google Search Ads **C++ library** by modularizing key components, writing unit tests, and redesigning public APIs to improve robustness and enforce correctness
- Wrote **technical design documents** for large-scale architecture changes and refactoring
- **Collaborated with client teams** to explore and enable new features with the core library
- Owned maintenance of a **multi-threaded** internal tool, debugging frontend and backend systems

AGS, LLC

Oct 2024 - Jan 2025

Software Engineer I - SDK

Duluth, GA

- Debugged issues and maintained casino cabinet video games using **C#** and **Unity**
- Applied **SCRUM** methodologies and ensured project stability through **code reviews** with the team
- Identified and resolved cabinet-specific **animation and graphical bugs** in the casino games
- Troubleshoot **network connections** between machines using **TCP** and **UDP protocols**

PROJECTS

Pokemon Showdown Machine Learning Bot written in Python using PPO

June 2024 - Jan 2025

- Trained a bot with a **Proximal Policy Optimization Algorithm** to play a turn-based battle
- Created a Python class to parse data from a **JavaScript datastream** that simulated the battles
- Utilized **2D arrays** to store information for both players' perspectives of the game

Portal Game developed in Unreal Engine 5 [\[LINK\]](#)

Feb 2024

- Solo-developed a 3D puzzle game with portals using Unreal Engine 5, Blueprints, and C++
- Used Blueprints and vector logic to create the **physics and rendering for portal travel** gameplay
- Created a **subsystem** using C++ to manage portal spawning and relocation

EDUCATION

Master of Science: Software Engineering

Dec 2023

San Jose State University

GPA: 3.39

Bachelor of Science: Computer Science

June 2020

University of California - Santa Cruz

GPA: 3.72